DocFlow AI - Technical Documentation

Frontend, backend, module architecture, setup commands, environment variables, and developer notes.

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Architecture Overview

DocFlow AI uses Django for the main application and a FastAPI microservice for AI document analysis. The system is modular, role-based, and organization-scoped.

Layer	Technology / Module
Backend	Django, Django templates, PostgreSQL, Redis/Celery optional, Django admin.
Frontend	Server-rendered HTML templates, Bootstrap, custom app.css, responsive sidebar/topbar.
AI Service	FastAPI service for extraction summary, category, risk detection.
Documents	Upload, validation, PDF text extraction, OCR fallback, versioning, AI fields.
Approvals	Requests, workflows, steps, actions, delegation, reminders, policies.
Audit	AuditLog model, helper logger, IP/user-agent metadata.
Reports	CSV/PDF/printable summaries and certificate generation.

Backend App Modules

App	Purpose
accounts	Custom user, profile, roles, login, organization user management.
organizations	Organization model, request numbering, profile/settings.
departments	Department CRUD and department head mapping.
approvals	Request lifecycle, workflow engine, actions, policies, delegation, reminders.
documents	Attachments, upload, OCR, AI processing, versioning.
audit	Audit logs and compliance tracking.
notifications	In-app and email notification support.
reports	Dashboards, CSV export, PDFs, printable pages.
dashboard	Role-wise dashboard pages and analytics.
superadmin	Platform-level organization/user management.

Frontend Documentation

The frontend is template-based. It uses base.html for authenticated layout, sidebar navigation, topbar, messages, and responsive mobile sidebar. Page templates extend base.html and use shared CSS classes.

Important templates

• templates/base.html - main authenticated layout.

• templates/approvals/request_detail.html - central request review page.

• templates/documents/attachment_detail.html - document metadata/OCR/AI/version page.

• templates/documents/attachment_upload.html - multiple file upload.

• templates/approvals/approval_inbox.html - approver action list.

• templates/dashboard/*.html - role dashboards.

• templates/reports/*.html - printable and report screens.

Important CSS classes

• page-header, page-subtitle, eyebrow for page headings.

• panel-card and panel-card-header for cards.

• stats-grid and metric-card for dashboard counts.

• modern-table for responsive tables.

• status-badge status-approved/status-rejected/status-review/status-draft for status labels.

• workflow-timeline and workflow-step-item for approval steps.

Backend Setup and Run Commands

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
python manage.py makemigrations
python manage.py migrate
python manage.py createsuperuser
python manage.py collectstatic
python manage.py runserver
```

FastAPI AI service

```
cd ai_service
uvicorn main:app --host 127.0.0.1 --port 8001 --reload
```

Celery optional

```
celery -A config worker -l info
celery -A config beat -l info
```

Environment Variables

Variable	Use
SECRET_KEY	Django secret key.
DEBUG	Development/production toggle.
DATABASE_NAME/USER/PASSWORD/HOST/PORT	PostgreSQL configuration.
AI_SERVICE_URL	FastAPI AI service base URL.
EMAIL_HOST/PORT/USER/PASSWORD	SMTP email notification setup.
REDIS_URL	Redis broker/cache URL.
OCR_ENABLED	Enable/disable OCR.
TESSERACT_CMD	Windows path to tesseract.exe.
POPPLER_PATH	Windows path to Poppler bin folder.

Developer Maintenance Notes

- Run python manage.py check after every module update.
- Run migrations after model changes.
- Do not add admin readonly fields that do not exist in the model.
- Keep URL names aligned with templates.
- Use select_related/prefetch_related for request detail pages.
- Keep organization isolation in every queryset.
- Keep audit logs for all important write actions.
- Before production, disable DEBUG and configure secure cookies/CSRF trusted origins.